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Task Description:

This joint project between the University of California, Irvine and Imperial College, London explores the use of large electric fields in combustion for the purpose of improving performance of energy conversion systems. The project this year focused on electrical effects that can be explored effectively in extended microgravity conditions and on the effects on flame saturation current of oxidizer dilution in the fuel stream. The studies included: 1) Continued numerical simulation of ion pathlines and ion-driven winds from small diffusion flames now extended to non-spherical flame shapes 2) Design of a new experimental configuration and procedures consistent with experiments that can be conducted in an extended zero gravity environment 3) Evaluation of how the flame saturation current changes as a function of air dilution into the fuel stream and whether this change can be used to characterize combustion performance
Task Description: The overall goal of this project is to understand the behavior of chemi-ionization in flames and to demonstrate the use of this understanding to manipulate combustion in beneficial ways (including enhancing flame stability and reducing flame soot). The project uses detailed measurements of electrical properties of flames in several ways.

First, we can create a neutral wind (ion driven convection) locally at the flame front; this is one of the rare and important action-at-a-distance effects on flames that can be used as a localized actuator. The effect of the wind has been demonstrated with a candle flame in parabolic flight based microgravity and in a spherical cage drop tower, but there has been no systematic study of the ion wind effect in the absence of the confounding effects of the approximately equal influence of buoyancy. The actuation of the flame can be used as part of a feedback control loop to maintain flame performance despite external fluctuations.

Second, we can use electric fields to affect soot formation and transport – ions and soot are closely linked because there are shared chemical pathways and because soot particles are charged by the flame ions. It is possible, therefore, to modify soot formation and to use charges attached to soot to affect soot transport. By adjusting the soot trajectory through the reaction zone it is possible to affect the oxidation of soot to prevent its eventual release or to modify the residence time of soot in different zones of the flame. It is also possible to use the neutral ion wind to change the mixing behavior of the flame and reduce soot in this way.

Finally, we can use the electrical properties to characterize the flame – there is a quantifiable relationship between the carbon content of a fuel and the ion current it can produce at saturation. This relationship is already exploited in a typical flame ionization detector (FID) analyzer for hydrocarbon emission where the carbon is burned in a hydrogen flame to create an ion current in proportion to the carbon influx. In direct flame interrogation, however, there are several confounding variables that can affect the ion current, including flame temperature (which generally affects the O atom concentration) and the reaction zone volume. Flame temperature can in turn be affected by levels of premixing, diffusion rates, fuel composition, and radiative loss. It is possible, therefore, to determine flame character from the ionization curve response.

In order to examine the above listed opportunities, we focused this year on two principal thrusts of the research. One focus was on the electrically controlled flame system, where experiments are conducted on a small diffusion flame sandwiched between two electrodes. With preparations for a flight experiment, we adjusted the laboratory experiment to mimic the constraints expected and we enhanced the numerical simulation to include flame shapes we expect in the microgravity environment. The second focus was on how to use the electrical properties of a flame to identify to identify its combustion behavior.

Task Significance: Understanding electric field interactions with flames is central to combustion control, particularly in situations near limit operation of the flame where small effects are amplified dramatically. Electric fields have been shown to modify burning rates as well as to enhance and reduce flame propagation even to the point of flame extinguishment. Our experiments provide a unique ability to make simultaneous measurements of both physical and electrical properties of flames.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Several of the applications we explore have potential uses in earth gravity as well as microgravity. For example, the application of ion winds for transport in miniature thermal systems can be valuable for heat transfer in small spaces. Understanding fundamental electrical properties of flames will also have contributions to earth-based combustion, particularly as regards soot formation and control, though the forces from electric fields will be less dramatic in the presence of natural convection.

Taking the three specific objective topics in turn, the following progress has been achieved over the past reporting period of the project –
Numerical Simulation of Small Flames in Electric Fields – Previous simulations focused on a spherical flame shape, characteristic of a neutrally buoyant flame. This flame shape is representative of ground-based experiments when the applied potential is large enough to partially balance buoyant natural convection. In microgravity, this flame shape is the base from where additional conditions will be explored. Recent simulations reflect the expected experimental setup on the ISS. In the interest of creating a data set that is representative of the relevant boundary conditions of the system, an important parameter to quantify would be the temperature near the burner tip.

Simulations on a 2D axisymmetric model of the flame in a downward driven configuration were run. The flame is assumed to have constant 1400 K temperature, with a flame height/width ratio of 3:1. The upper mesh has been removed to match our current experimental embodiment and we have assumed that the ion concentration is independent of flame shape. Our experiment has been altered so that polarity is set relative to a grounded capillary (active mesh configuration) as opposed to a grounded mesh. Simulations have shown that by applying a large negative potential at the lower mesh relative to the capillary, the potential distribution local to the capillary and the resulting path lines are equivalent to our original configuration. The potential distribution is slightly altered in the far field due to the experimental enclosure being essentially at ground. This change would potentially aid in ion collection at the lower plane by effectively limiting the number of surfaces which the ions would be drawn to.

The results show that the outermost ion path lines for a downward driven flame are within a 3 cm radius about the capillary. The surface profile shows that heat is drawn downward as expected from our laboratory-based experiments. The ion wind effect is the dominant fluid flow mechanism as opposed to buoyancy driven natural convection. This is

shown by the direction of the arrows. To quantify the collected ion current, total ion flux at the charged mesh is determined as a function of radial distance from the axis of symmetry. Taking the surface integral and scaling according to ion concentration and charge, results in a current of 0.28 microA, which is on the order of our experimental values. The ion current is approximately two-thirds that found in the spherical flame simulations. Future work planned for this aspect of the project includes taking schlieren images of the downward driven flame and recreating the dimensions in our model to further validate the simulation.

Conceptual Design of Flight Compatible Experiment – Design of the extended microgravity spaceflight experiment (now called E-FIELD — Electric-Field Effects on Laminar Diffusion Flames) is a coordinated effort among NASA engineers and the research group at the University of California, Irvine. The experiment is currently in a preliminary design state, and is expecting a critical review early in 2008. The design aspects completed thus far include burner selection, conceptual drawings for an experimental configuration within the test chamber, electrical schematics for electric field generation and ion current measurement, an initial test matrix, and experimental requirements. The following is a brief discussion of each of these design aspects and the laboratory-based experiments performed to support some of the design decisions. The first E-FIELD burner proposed was a spherical, porous steel burner in order to limit the amount of time that ISS mission specialists would have to spend reconfiguring the experiment on-orbit. Several laboratory experiments were conducted to determine if the spherical burner was compatible with our spaceflight experimental objectives. These experiments showed that the ion-driven wind was not effective in substantially modifying combustion in this burner. Since observing flame shape changes in the high-voltage electric field was an important experimental objective, the proposed spaceflight experiment will employ the small gas jet flame used in the prior laboratory experiments.

Although the spherical burner will not be used in the E-FIELD experiment, the laboratory experiments were indirectly responsible for the development of the high-voltage protection circuit now implemented in the E-FIELD design. During one experiment, a high-voltage arc formed between the burner and the mesh due to breakdown in the air and the electrical pathway created by the mobile ions. This arc created a short circuit current that destroyed the data acquisition hardware used to measure the ion current. To prevent this from happening in the laboratory and spaceflight experiments, a Zener diode was placed in parallel with the sense resistor. This protection circuit tested successfully in all experimental configurations. In addition, it was determined that the Zener diode does not interfere with the ion current measurements so long as the cathode direction is flipped when altering the polarity of the high-voltage mesh.

The test matrix for the experiment design requires disclosure of fuel types, flow rates, instruments required, and the operational sequence for those instruments. One important experimental sequence for E-FIELD is capturing the voltage-current characteristic (VCC) curve for various fuels and mixtures. Several experiments were conducted to determine the requirements for sampling rate, sampling period, and number of voltage steps required to obtain the VCC curves. It was observed that longer sampling periods resulted in a higher current/volt ratio in the super-saturated region of the VCC curve. This is because the overall settling time for the flame exceeds the initial electrical response predicted in simulations and observed in previous step-response experiments (approximately 0.25 seconds or less). After the initial step-response settling, the ion current increases until it reaches a steady-state value. This slow timescale may be due to thermal interactions with the burner tip, which is allowed to cool when the flame is pulled away from the burner tip by the ion wind force at mesh potentials greater than 5 kV in the positive mesh polarity configuration, and 2 kV in the negative mesh polarity configuration. In order to capture the fully settled ion current value, the sampling rate must be longer than the flame settling time. The conclusion from the experiments was that the maximum ion current is obtained at 5-second sampling periods. Sampling periods greater than 5 seconds result in the same curve, and sampling periods below 5 seconds result in a smaller current/volt ratio in the super-saturated region. We expect less thermal interaction at zero gravity since the flame will not ever burn around the gas jet, but in order to conserve fuel on the spaceflight experiment, the number of data points must be taken at a resolution that still enables the saturation point to be identified.

Task Progress:

Characterizing Flames by their Saturation Currents – Saturation current densities (js) for premixed flames of numerous hydrocarbons with air have been documented as functions of fuel-air ratio and the initial mixture temperature, as long ago as 1964 (J. Lawton and F.J. Weinberg, Proc. Roy. Soc. A.277, 468, 1964; see also "Electrical aspects of combustion" Clarendon Press, Oxford, 1969). However, the mixture compositions did not extend much beyond stoichiometric into the fuel-rich region because the accuracy of the flat-flame method used would have been compromised by sooting and by the formation of a secondary diffusion flame due to the burning of excess fuel. In these prior studies, the dependence on flame temperature could be represented by an activation energy - such that an increase of 100K in flame temperature would result in a 2 - 4 fold (depending on the hydrocarbon) increase in ionisation rate. At the other extreme, total saturation currents, Js, measured for pure diffusion of flames of hydrocarbons and including inert diluents are also available in the literature (e.g., D. Boothman, J. Lawton, S.J. Melinek and F.J. Weinberg, Proc. of The Combustion Institute, vol. 12, p. 969, 1969). These total saturation currents indicate an approximately constant yield of ions per carbon atom and, by contrast with premixed flames, are largely independent of temperature and diluents. “Diluents” here include hydrogen which, although it is a fuel, does not contribute to any ionisation. This relationship between carbon atom flux and ion current has been used as the basis for quantitative detectors, e.g., in gas flame ionization detectors (FID) for hydrocarbon analysis in combustion exhaust. In between the compositions described above there is an extensive unexplored range of fuel-rich mixtures extending from small additions of air to a fuel jet all the way to beyond the upper limit of flammability at about 85% of air.

In the experimental system, volumetric flow rates of fuel (Qf), which was natural gas throughout this study, and air (Qa) were metered by rotameters calibrated against bubble meters. The diameter of the outer tube is 25mm. The polarity of the burners was always negative with respect to ring electrode, to minimize wind effects by causing the low-mobility positive ions to travel the shorter path to the burner. Even though some of the electrons are likely to attach during their longer travel and become negative ions, the predominant ion wind was directed downwards (see below). Different burner diameters were used because each was restricted to a limited operability range in terms of flow rate and fuel/air ratio. In the time available, 1, 3 and 4 mm diameters were explored. The operability limits were due to: 1) sooting in the case of diffusion flames, at high flow-rates. The limit was set arbitrarily at luminous lengths of less than 10% of the total flame length; 2) blow off; 3) flashback for premixed flames; 4) excessive flame distortion by applied fields. An extreme case occurred for very fuel rich flames in large fields. Positively charged soot particles can be seen looping back to deposit on the burner and inducing a toroidal vortex in the flow; and 5) the absence of a saturation plateau at high flow rates when the rate of ion generation always exceeds the rate of the removal by the field.

A summary of these limitations are mapped; we find that for premixed flames - in particular compositions lean of stoichiometric - are attainable only for the larger burner diameters. The effect of preheating was also briefly investigated for the 1 mm diameter burner with a heating coil wound around its mouth.

The experimental results showed that there was no contribution to the current at any voltage (V) from “clean” burners. For soot-coated burners and for electrically heated burners, some field emission set in at high potentials - well beyond values at which saturation current measurements were completed. In many cases there was a very gradual increase in current beyond the initial plateau in the current/voltage relationship. Since this must be due to some modification of the flame by the field, no matter how small, the value of the saturation current was always taken at the onset of the plateaux. In the above-mentioned case of large currents - for premixed flames at large flow rates, dJ/dV never tended to zero and they were considered to be outside the tractable regime. The limits to this regime can be plotted on a fuel-flow vs. air-flow diagram so that loci of constant compositions are straight lines radiating from the origin, while total flow rate loci linearly join the corresponding values on each axis. We find that the locus of stoichiometric compositions is, to a good approximation, the line bisecting the angle between the two limits of flammability. Only flames stabilizable on the 4mm diameter burner penetrate into the fuel-lean domain. With the exception of the preheat case, all the results can be shown as saturation current plotted against total flow (Q) to emphasize the different ranges of the three burners. In this kind of mapping, however, the fuel air ratio, of course also varies; this is discussed in the next section. The peaks correspond to stoichiometric conditions. It was not feasible to plot a three-dimensional graph because of the large variations involved.

From previous work, outlined earlier, plotting (Js/Qf) (i.e., effectively the number of ion pairs per carbon atom) against the proportion of air added, should at least start as a horizontal line. According to prior studies in the literature the magnitude should be approximately 0.035 microA for methane, though some variations were seen in the experimental value, depending on the type of diffusion flame. This relationship would not be expected to extend into the premixed regime, in any case. For premixed conditions we can roughly estimate what would be expected from the previously mentioned accurate measurements of saturation current density, js, for flat un-stretched flames, since Js should equal A* js, where A is the area of the premixed flame. Since, according to the simple theory of premixed flames, the total volumetric flow rate, Q = A*Su , where Su is the normal burning velocity, we predict that Js =Q*js/Su. When all the total flow rate results are replotted in this format, it is immediately apparent that this is a rational basis for correlating the wide range of results. The larger burners give close to the predicted results at the few points where comparisons are possible. The 1mm burners tend to high values at the diffusion flame end, though it should be pointed out that a change in flow metering equipment was necessitated by the almost 10 fold change in the flow rate ranges. What seems surprising is that admixture of air to a pure fuel gas, even in amounts of up to 70%, appears to be no different in its effect on ionisation than dilution by an inert. It is not until the air content reaches the upper (fuel-rich) limit of flammability that there is a remarkable rise in charge per carbon atom. There is an inner cone visible at this composition, but there is no conspicuous difference in its appearance from that beyond this limit on the fuel-rich side. For the one set of results approaching the lower limit of flammability, the saturation current also falls below its diffusion flame value; the value is close to the prediction from our Js =Q*js/Su calculation. Another intriguing feature, which should provide a challenge for numerical simulation, is that all those burners capable of stabilising premixed flames appear to show a dip in Js/Qf near the rich limit of flammability, just before the rise.

Finally, there was time for just one set of measurements on the effect of raising the initial temperature to about 320C for the 1 mm burner. The result is exactly in accordance with the aforementioned principles: there is no discernible effect on the diffusion flame region of the graph which again extends to about 70% air admixture. Beyond that, however, it is now possible for the first time to reach into the premixed regime with a 1mm burner, with the concomitant rise in in Js/Qf.

Bibliography Type:	Description: (Last Updated: 11/26/2025)
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